

Kubernetes Architecture

Notes Summary

1. Master Node / Control Plane

Master Node controls and manages the entire Kubernetes cluster.

Components:

API Server

The API Server is the central communication point of the Kubernetes cluster.

- Receives requests from users and other components.
- Allows Kubernetes components to communicate with each other.
- Also handles authentication, authorization and admission control.

etcd

etcd is a distributed key-value database that stores the cluster state and configuration information.

- Stores important cluster information.
- Acts like the database of Kubernetes.

Scheduler

The Scheduler decides which worker node should run a newly created Pod.

- Checks available CPU and memory.
- Selects a suitable worker node for the Pod.
- Also considers taints, tolerations, and node/pod affinity rules.

Controller Manager

The Controller Manager manages different controllers and ensures that the actual cluster state matches the desired state.

- Monitors the cluster.
- Maintains the desired state.
- Manages different Kubernetes controllers.

Cloud Controller Manager (CCM)

The Cloud Controller Manager connects the Kubernetes cluster with the underlying cloud provider's API.

- Manages cloud-specific resources such as load balancers, storage, and networking.
- Separates cloud provider logic from the core Kubernetes components.
- Runs cloud-specific controllers like the node, route, and service controllers.

Master Node Easy Revision

API Server	Communication
etcd	Stores cluster data

Scheduler	Selects worker node
Controller Manager	Maintains desired state
Cloud Controller Manager	Connects to cloud provider

2. Worker Node

Worker Node is the machine where Kubernetes applications and Pods actually run.

Components:

Kubelet

Kubelet is an agent that runs on each worker node and ensures that the assigned Pods are running and healthy.

- Communicates with the API Server.
- Manages Pods on the worker node.
- Works with the container runtime.

Kube-proxy

Kube-proxy maintains network rules and helps route network traffic to Kubernetes Services and Pods.

- Handles network rules.
- Helps route traffic to the correct Pod.

Container Runtime

Container Runtime is the software responsible for running and managing containers.

- Examples: containerd, CRI-O
- Note: Docker was used earlier, but Kubernetes removed direct Docker support (dockershim) from v1.24 onward.